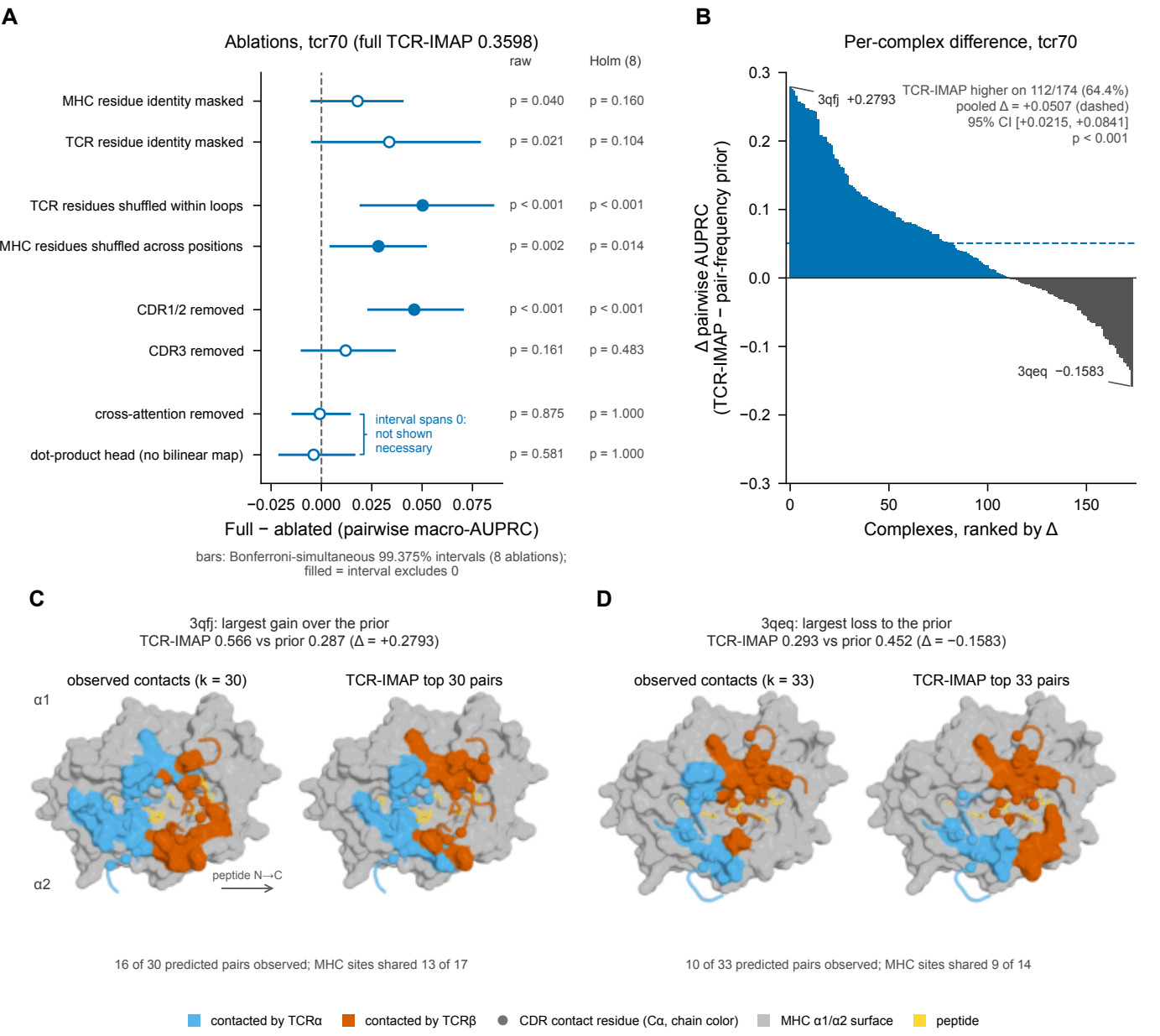

A
Ablations, tcr70 (full TCR-IMAP 0.3598)
raw
Holm (8)
MHC residue identity masked
p = 0.040
p = 0.160
TCR residue identity masked
p = 0.021
p = 0.104
TCR residues shuffled within loops
p < 0.001
p < 0.001
MHC residues shuffled across positions
p = 0.002
p = 0.014
CDR1/2 removed
p < 0.001
p < 0.001
CDR3 removed
p = 0.161
p = 0.483
cross-attention removed
p = 0.875
p = 1.000
interval spans 0:
not shown
necessary
dot-product head (no bilinear map)
p = 0.581
p = 1.000
−0.025 0.000 0.025 0.050 0.075
Full − ablated (pairwise macro-AUPRC)
bars: Bonferroni-simultaneous 99.375% intervals (8 ablations);
filled = interval excludes 0
B
Per-complex difference, tcr70
TCR-IMAP higher on 112/174 (64.4%)
pooled Δ = +0.0507 (dashed)
95% CI [+0.0215, +0.0841]
p < 0.001
3qfj +0.2793
3qeq −0.1583
Δ pairwise AUPRC
(TCR-IMAP − pair-frequency prior)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
50
100
150
Complexes, ranked by Δ
C
3qfj: largest gain over the prior
TCR-IMAP 0.566 vs prior 0.287 (Δ = +0.2793)
observed contacts (k = 30)
TCR-IMAP top 30 pairs
α1
α2
peptide N→C
16 of 30 predicted pairs observed; MHC sites shared 13 of 17
D
3qeq: largest loss to the prior
TCR-IMAP 0.293 vs prior 0.452 (Δ = −0.1583)
observed contacts (k = 33)
TCR-IMAP top 33 pairs
10 of 33 predicted pairs observed; MHC sites shared 9 of 14
contacted by TCRα
contacted by TCRβ
CDR contact residue (Cα, chain color)
MHC α1/α2 surface
peptide